

Evaluation of Cytotoxicity and Flavonoid Presence of Cabbage Extract (*Brassica oleracea*)

¹J.C.E. Cheng Roa, ²J.J. Evora, ³J.J. Lopez, ⁴J.M. Marcelino Jr., ⁵S.K.B. Relayo

^{1,2,3,4,5}Group 8, Grade 12 – STEM 1, Integrated School

^{1,2,3,4,5}De La Salle Araneta University, Don Salvador Araneta Campus, Victorina Avenue, Malabon City
1475, Metro Manila, Philippines

¹john.chengroa@dlsau.edu.ph, ²jerome.evora@dlsau.edu.ph, ³justine.lopez@dlsau.edu.ph,

⁴jefferson.marcelino@dlsau.edu.ph, ⁵samantha.relayo@dlsau.edu.ph

ABSTRACT

The main objective of this study was to extract cabbages using microwave-assisted extract, followed by qualitative screening for the presence of flavonoids and assessing cytotoxicity of the extract. The study created 250 mL cabbage extracts at 50%v/v concentration using microwave-assisted extraction, tested for flavonoids using Alkaline Reagent Test and Concentrated Sulfuric Acid Test, and evaluated the cytotoxicity using *Allium cepa* Lethality Assay. Alkaline reagent phytochemical screening of cabbage extract showed strong presence of flavonoids, while concentrated sulfuric acid phytochemical test showed minimum presence of flavonoids. The *Allium cepa* lethality assay results showed that cabbage extract with the highest concentration (50%) had the least root length yet had no inhibitory effect on root growth. Which concludes that cabbage extract showed presence of flavonoids and is not cytotoxic. The study recommends an efficient timeframe and multiple *Allium cepa* replicates.

Keywords: Cabbage Extract, Flavonoids, Microwave-Assisted Extraction

INTRODUCTION

Existing research from food and science technology international shows a result that cabbage is a rich source of bioactive compounds that contains antibacterial and antioxidant properties (Jaiswal, et al., 2013). Cabbage extract is also known as cabbage water extracted this way, can be consumed, while retaining beneficial phytochemicals (Begum, 2023). In which Kaempferol, quercetin, and apigenin were the major flavonoids existing in cabbages except for raw and pickled red cabbages, where cyanidin was the predominant flavonoid

(Chun et al., 2014). Many studies have suggested that Flavonoids exhibit biological activities, including antiallergenic, antiviral, anti-inflammatory, and vasodilating actions. However, most interest has been devoted to the antioxidant activity of flavonoids, which is due to their ability to reduce free radical formation and to scavenge free radicals. (Pietta, 2010)

Microwave-Assisted Extraction is based on electromagnetic waves that cause the cell structure to change (Nour et al., 2021). By applying “micro-waves” to the extract, the microwave radiation heats the solvent and the

cabbage tissue. This heat causes thermal expansion, which causes the cell wall components to expand and weaken. In some cases, direct rupture. The intense microwave energy can directly rupture the cell walls. This extraction method is more efficient because it works much quicker than heating the solvent directly.

Allium cepa test is a valuable tool for screening substances for their potential to cause genetic damage and cell death (Martins, 2021). The test is based on the principle that exposure to genotoxic or cytotoxic substances can cause root growth inhibition and chromosomal aberrations in onion root tip cells. It is a rapid, inexpensive, and easy-to-perform test that can be used in a variety of settings (Bonciu et al., 2018).

This research aims to obtain cabbage extract using microwave-assisted extraction, followed by qualitative screening for the presence of flavonoids and assessing cytotoxicity of the extract via *Allium cepa* test.

Assessing the lethality and presence of flavonoids in cabbage solution that was extracted using microwave-assisted extraction has come up with the following

problems. Does the cabbage extract exhibit flavonoid presence? and does the cabbage extract exhibit cytotoxic effects on red onion (*Allium cepa*) root tips compared to the control group exposed to distilled water?

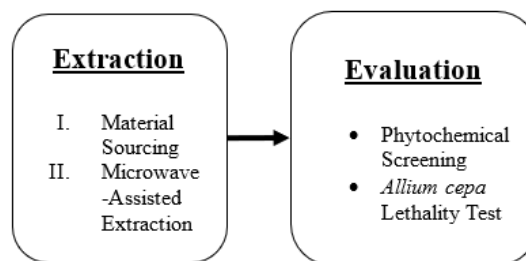
This study has potential for the determination of the use of cabbage extracts for stem cell treatments, due to its influence of cell behavior. It has an impact as fertilizer and for plant growth. Like improving the growth, yield of plants and extending the shelf-life of post-harvested plants that have a high decay rate.

METHODS

This study utilizes experimental research design because it manipulates the cabbage extract concentration and measures the *Allium cepa* root growth and flavonoid presence. The study was conducted at the DLSAU LSB Laboratory, where the room temperature averaged from 25-30°C.

Figure 1

Conceptual Framework



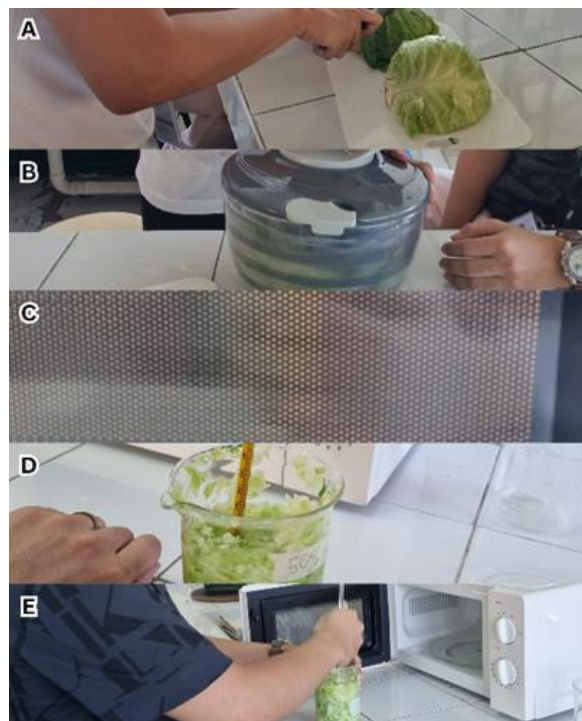
Material Sourcing

The study uses 1 cabbage head, 2L distilled water, a microwave oven that can reach 99.85°C, 500mL beakers, 300mL glass jar, a watch glass, a glass rod, thermometer, small cooler with ice, knife and cutting board, and a vegetable spinner. NaOH, HCl, and H₂SO₄ were used for Phytochemical Screening, along with 3 test tubes, test tube rack, and a pipette + aspirator. 10 plastic bottles, cutter, 5 Red Onions (*Allium cepa*) for the cytotoxicity assay.

Microwave-Assisted Extraction

Figure 2

Microwave-Assisted extraction process



Section **A** shows the cabbages that were cut into small pieces and distilled water are to be distributed as 125g cabbage to 250mL water for a 50%*m/v* concentration. Section **B** shows that the cabbages were washed with distilled water and dried using a vegetable spinner. Section **C** shows the extraction of the cabbage solution for 45 seconds, wherein the extraction process should last for only 25 minutes only (Xu et al., 2021) and approximately 90°C should be the upper limit of the extract's temperature (Pimentel-Moral et al., 2018). Section **D** shows the measurement of the solution's temperature after 45 seconds in the microwave. Section **E** shows that the cabbage solution was mixed thoroughly for 1-2 minutes, which varies depending on the temperature rising too fast. Section **C, D, E** were repeated until the 25-minute time limit was reached.

Phytochemical Screening

An *in vitro* qualitative test assessed flavonoid presence in the extract. Two methods were used. Alkaline Reagent Test, wherein 2-3 drops of 2M sodium hydroxide (NaOH) solution were added to 2 mL of extract. A yellow color turning colorless with diluted HCl indicates flavonoids. Another test was the Concentrated Sulfuric Acid Test

(Tyagi, 2017), wherein adding concentrated sulfuric acid to the extract and observing an orange color suggests flavonoids.

Figure 3

Qualitative Phytochemical Screening Reference



ALKALINE REAGENT

The test tube on the left shows the addition of NaOH, while the test tube on the right shows the addition of HCl.



CONC. H₂SO₄

The test tube shows the addition of H₂SO₄

***Allium cepa* Lethality Test**

Five red onion bulbs were grown in water for 24-48 hours to develop roots (1-2 cm). Next, 300 mL solutions were prepared: distilled water (control) and 500 mg/mL cabbage extract diluted to 5%, 10%, 20%, and 50%. Each onion was placed in a solution and observed for 5 days.

Figure 4

Allium cepa lethality assay set-up



Root growth was compared to control. If root growth was inhibited (fewer roots

compared to control), the extract concentration was considered effective. Otherwise, similar root growth to the control indicated no effectiveness.

Figure 5

Measurement of Allium cepa after 5 days of treatment



RESULTS

The cabbage extract with a 50% m/v concentration was extracted using microwave-assisted extraction for 25 minutes averaging a temperature of 63.83°C for about 6 cycles of microwaving, measuring, and mixing.

For alkaline reagent test, the researchers added 2 drops of 2M sodium hydroxide (NaOH) into 2 test tubes containing 2mL of the extract, while adding 2 drops of 6M of HCl in one test tube. For the concentrated sulfuric acid test, the researchers added 6M of H₂SO₄ into 2mL of extract. Expected results were marked as (+), while results that

differ, or no results were marked as (-). Table 1 shows the results of the phytochemical screening for flavonoids of the cabbage extract:

Table 1
Results of Qualitative Phytochemical Screening

Tests	Observations	Presence ^[a]
Alkaline Reagent	Intense Yellow, Colorless when HCl is added	+
Coc. H ₂ SO ₄	Slight Brown	-

a. (+) = present, (-) = absent

Alkaline reagent qualitative phytochemical screening showed strong presence of flavonoid, with an intense yellow color after NaOH was added and it faded into colorless after a drop of HCL was added. However, the concentrated sulfuric acid test showed minimal to no presence of flavonoids, as it faded into light brown color instead of the expected blue color.

The researchers diluted the 50%v/v cabbage extract into five (5) separate solutions of 5%, 10%, 20%, 50% and a control solution without any cabbage extract, which was prepared into five (5) 500mL plastic bottles. Then 5 red onions were added into another five (5) 500mL plastic bottles containing distilled water, which only the roots were exposed to. For 5 days, the red onions were

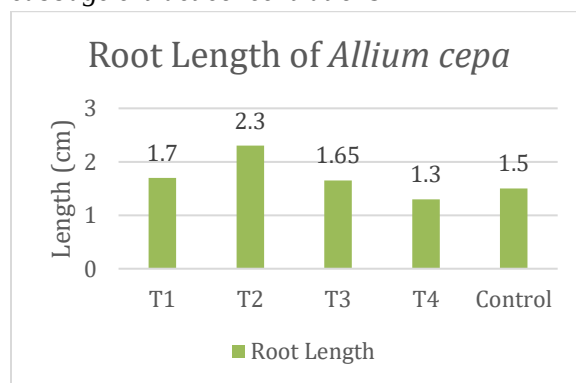
exposed to the extracts with different concentrations. Table 2 shows the growth of the red onion roots, measured in centimeters (cm) at day 5:

Table 2
Root growth of red onions exposed in cabbage extracts with different concentrations at day 5

Concentration	Root Length (cm)
T ₁	1.7 cm
T ₂	2.3 cm
T ₃	1.65 cm
T ₄	1.3 cm
Control	1.5 cm

Figure 6 shows the bar graph of the root growth of the red onions that was exposed to the different concentrations of cabbage extract:

Figure 6
Root growth of red onions exposed to different cabbage extract concentrations



The lower the value of the root length, the greater the cytotoxicity.

Root length of the treatment groups ranged from 1.3 cm to 2.3 cm, while control root length was 1.5cm. The highest concentration (T₄) showed the shortest *Allium cepa* root length, making it most effective against the rest of the concentration groups.

DISCUSSIONS

Phytochemical screening results from alkaline reagent test and concentrated sulfuric acid test align with the study of Tarigan (2022) who stated cabbage extract contained flavonoids which had potential as an antioxidant. Since flavonoids are often associated with antioxidant activity in plants.

While the *Allium cepa* lethality test result slightly differed from the study of Farag (2010) who stated green cabbage extract exhibited a pronounced cytotoxic effect. This difference could be due to several factors which includes the cabbage age which older cabbages might have different phytochemicals or bioactive compounds present, extraction method that might affect the extracted compounds necessary for the cytotoxicity, and the concentration that was used from the study of Farag (2010).

Another study of Abu (2024) who conducted a quantitative analysis on the

cytotoxicity of green cabbages shows that the result obtained showed that the cabbage extract had a CC₅₀ value of 10.6 µg/mL, while vincristine (standard) had an IC₅₀ of 0.30 µg/mL. Note that the lower the value, the higher the cytotoxicity. This aligns with our results, as this shows lower levels of cytotoxicity. In which, the control exhibited greater cytotoxic properties than the cabbage extract.

Cabbage extract showed effects on cytotoxicity and presence of flavonoids. The Alkaline Reagent Test showed strong presence of flavonoids, while concentrated sulfuric acid showed weak to no presence of flavonoids. The *allium cepa* lethality test showed that T₄ had the greatest effectiveness, while T₁ to T₃ showed worse results than Control.

Based on the findings of the study, the following conclusions are drawn. Firstly, cabbage extract showed the presence of flavonoids. Lastly, cabbage extract was not cytotoxic.

The following recommendations are made for future studies. Firstly, conduct the phytochemical tests and cytotoxic assay within one week if using cabbage solution

extracted via M-AE. Secondly, consider adding multiple replicates for the cytotoxic assay which allows for data analysis like ANOVA. Lastly, Change the concentration of the extract wherein the initial concentration of the solution before the extraction phase should be greater than 60% m/v to determine proper cytotoxicity.

REFERENCES

Abu T., Ezea B.O., Yusuf S.A., Bamidele O.O., Sidiq L.O., Yusuf A.O., Ogbole O.O. (2024, May 9). Phytochemicals, proximate and elemental analysis, antioxidant and cytotoxic potentials of purple (*Capitata F. rubra*) and green (*Capitata Linn.*) *Brassica oleraceae* (cabbage). *Journal of Pharmacy & Bioresources*, Vol. 21 (No. 2).
<https://www.ajol.info/index.php/jpb/article/view/270166/254997>.

Begum J. (2023). Health Benefits of Red Cabbage.
<https://www.webmd.com/diet/health-benefits-red-cabbage/>

Bonciu E., Firbas P., Fontanetti C.S., Jiang W., Karaismailoglu M.C., Liu D. (2018, Sep 14). An evaluation for the standardization of the *Allium cepa* test as cytotoxicity and genotoxicity assay.

<https://www.tandfonline.com/doi/full/10.1080/00087114.2018.1503496>.

Chun, O., Smith, N., Sakagawa, A., & Lee, Y. (2014). Antioxidant properties of raw and processed cabbages. *PubMed National Library of Medicine*.
<https://pubmed.ncbi.nlm.nih.gov/15223595/#:~:text=Kaempferol%2C%20quercetin%2C%20and%20apigenin%20were,%2D117.7%20mg%2Fkg>).

Farag M.A., Motaal A.A.A. (2010, Feb 23). Sulforaphane composition, cytotoxic and antioxidant activity of crucifer vegetables.
<https://www.sciencedirect.com/science/article/pii/S2090123210000081>

Jaiswal, et al. (2013, September 27). Application of Baranyi function to model the antibacterial properties of solvent extract from Irish York cabbage against food spoilage and pathogenic bacteria.
<https://journals.sagepub.com/doi/abs/10.1177/1082013211399667>

Martins J.G. (2021, Nov 23). Screening do potencial hidrogeniônico toxicidade em *Allium cepa* de infusões populares frente a urolitíase por metodologia analítica.
<https://typeset.io/papers/screening-do->

[potencial-hidrogenionico-e-toxicidade-em-allium-sugk8zit8i.](#)

Nour, A. H., Oluwaseun, A. R., Nour, A. H., Omer, M. S., & Ahmed, N. (2021, February 23). Microwave-assisted extraction of bioactive compounds (review). <https://www.intechopen.com/chapters/75284>

Pietta, P.G. (2010). Flavonoids as Antioxidants.

<https://doi.org/10.1021/np9904509>

Pimentel-Moral, S., Borrás-Linares, I., Lozano-Sánchez, J., Arráez-Román, D., Martínez-Férez, A., and Segura-Carretero, A. (2018). Microwave-assisted extraction for Hibiscus sabdariffa bioactive compounds. J. Pharm. Biomed. Anal. 156, 313–322. doi: 10.1016/j.jpba.2018.04.050

Tarigan L.D.B., Lister I.N.E., Fachrial E., Armansyah A. (2022, Feb 22). Phytochemical of Ethanolic Extract of Cabbage (Brassica oleracea) Collected from Medan, Indonesia. <https://talenta.usu.ac.id/JCNaR/article/download/10207/5598#:~:text=The%20test%20results%20showed%20that,and%20steroid%20or%20triterpenoid%20compounds>

Tyagi, T. (2017). PHYTOCHEMICAL SCREENING OF ACTIVE METABOLITES PRESENT IN EICHHORNIA CRASSIPES (MART.) SOLMS AND PISTIA STRATIOTES (L.): ROLE IN ETHANOMEDICINE. Asian Journal of Pharmaceutical Education and Research, 6(4), 40–56. http://ajper.com/admin/assets/article_issue/1508170696.pdf

Xu J., Wu J., Qi J., Li J., Liu Y., Miao Z., Qiu G., and Jia W. (2021). "Microwave-assisted extraction of flavonoids from Phyllostachys heterocycla leaves: Optimization, mechanism, and antioxidant activity in vitro," Bioresources 16(4), 8060-8081. <https://bioresources.cnr.ncsu.edu/resources/microwave-assisted-extraction-of-flavonoids-from-phyllostachys-heterocycla-leaves-optimization-mechanism-and-antioxidant-activity-in-vitro/#:~:text=The%20optimal%20microwave%2Dassisted%20extraction,microwave%20power%20of%20559%20W>

APPENDICES

Appendix A

Microwave-assisted extraction summary log

Cycle Count	Extraction Time	Temperature	Cooldown
1	45 seconds	39°C	1 minute
2	45 seconds	48°C	2 minutes
3	45 seconds	66°C	2 minutes
4	45 seconds	65°C	2 minutes
5	45 seconds	71°C	2 minutes
6	45 seconds	76°C	2 minutes

Appendix B

Microwave-assisted extraction documentation



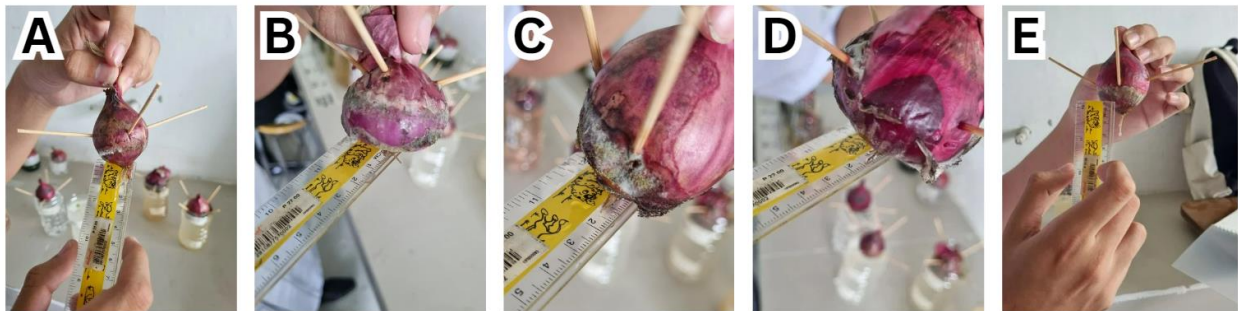
Appendix C

Allium cepa lethality test documentation



Appendix D

Allium cepa measurements on day 5



Sample A was T_1 measuring 1.7cm, **Sample B** was T_2 measuring 2.3cm, **Sample C** was T_3 measuring 1.65cm, **Sample D** was T_4 measuring 1.3cm, **Sample E** was Control measuring 1.5cm.